ECEN 5692 HW 8 Matt Hartnett

Problem 2.2.A.2:

1) BPSK: $S = \{m : m \in \{-1, 1\}\}$

$e_{x(t)} = E[|X|^2]$

$= E[\int |x(t)|^2 dt]$

$= E[\int |\Sigma_k X_k u_k(t)|^2 dt]$

$= E[\int \Sigma_{k_1} X_{k_1} u_{k_1}(t) \Sigma_{k_2} X^*_{k_2} u^*_{k_2}(t) dt]$

$= E[\Sigma_{k_1} \Sigma_{k_2} X_{k_1} r_u(k_1, k_2) X^*_{k_2}]$

$= \Sigma_{k_1} \Sigma_{k_2} r_u(k_1, k_2) E[X_{k_1} X^*_{k_2}]$

Since $x \leftarrow S$ is uniformly distributed, $E[X_{k_1} X^*_{k_2}] = e_s \delta[k_1 - k_2]$

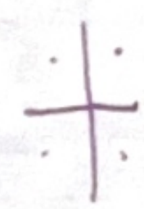
$e_{x(t)} = \Sigma_{k_1} \Sigma_{k_2} r_u(k_1, k_2) e_s \delta(k_1 - k_2)$

$= e_s \Sigma_{k_1} e_{u_k(t)}$

$e_s = E[\|X_k\|^2] = a^2$ $\quad P_{X_k}(a) = P_{X_k}(-a) = 0.5$ for BPSK

$\boxed{e_{x(t)} = 1, \ a = 1}$

2) QPSK: $S = \{m_I + j m_Q : m_I, m_Q \in \{-1, 1\}\}$



~~$E[|X_k|^2] = 1$ for all values of k~~

~~$\boxed{e_s = 1, \ a = 1}$~~

$|X_k| = \sqrt{1^2 + 1^2} = \sqrt{2}$ $\quad |X_k|^2 = 2$ for all values of k

$E[|X_k|^2] = 2$

$e_{ax(t)} = E[|aX_k|^2] = |a|^2 E[|X_k|^2]$

$1 = |a|^2 2$

$a = 1/\sqrt{2}$

$\boxed{e_{x(t)} = 2, \ a = 1/\sqrt{2}}$

3) 8-PSK: $S = \{e^{jm2\pi/8} : m \in \{0, 1 \ldots 7\}\}$

$|e^{jm2\pi/8}| = 1$ for all values of m

$\boxed{e_{x(t)} = 1, \ a = 1}$

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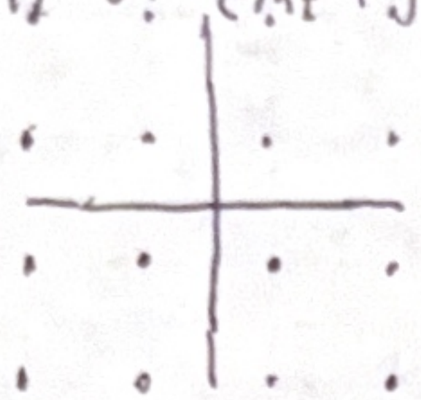
Problem 2.2.A.2 (cont):

4) $S: \{m_I + j m_Q : m_I, m_Q \in \{-3,-1,1,3\}\}$



$$|X_m| = \begin{cases} \sqrt{2} & \text{for } |m_I|=1,\ |m_Q|=1 \\ \sqrt{10} & \text{for } |m_I|=1 \text{ \& } |m_Q|=3 \ \ |\ |m_I|=3, \text{ \& } |m_Q|=1 \\ 3\sqrt{2} & \text{for } |m_I|=|m_Q|=3 \end{cases}$$

$$E[|X_m|^2] = 1/16\left(4(\sqrt{2})^2 + 8(\sqrt{10})^2 + 4(3\sqrt{2})^2\right)$$

$$= 1/16\,(8 + 80 + 72)$$

$$e_{x(t)} = 10$$

$$e_{ax(t)} = 1 = E[|a\,x(t)|^2]$$

$$1 = |a|^2 E[|x|^2] \Rightarrow a = 1/\sqrt{10}$$

$$\boxed{e_{x(t)} = 10,\ a = 1/\sqrt{10}}$$

Problem 2.2.A.3:

$X(t) = \sum_{l=1}^{n} X_l\, u(t - l t_b)$, $e_s = 1$, $1/t_b = 5\,MHz$, $u(t) = 1/\sqrt{t_b}\ \mathrm{sinc}(t\, t_b)$

$$PSD_{x(t)}(f) = \frac{1}{\text{duration of } x(t)} E[|\mathcal{F}(x(t))(f)|^2]$$

duration of $x(t) = n \cdot t_b$

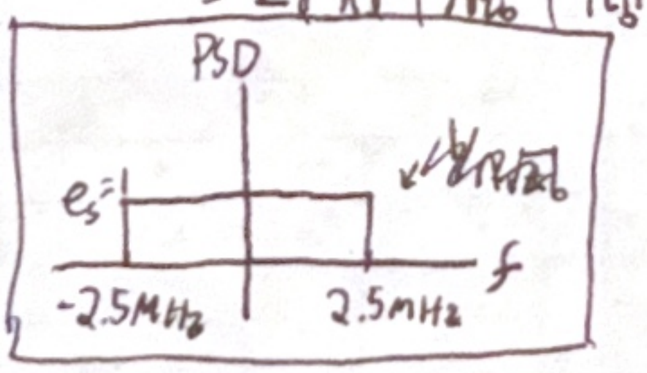
$$\mathcal{F}(x(t)) = \mathcal{F}\left(\sum_{l=1}^{n} X_l\, u(t - l t_b)\right)$$

$$= \sum_{l}^{n} X_l\, \mathcal{F}(u(t - l t_b))$$

$$= \sum_{l}^{n} X_l \left(1/\sqrt{t_b}\left(|t_b|\ \mathrm{rect}(f t_b)\right)\right)$$

$$PSD = \frac{1}{n t_b} \sum_{l=1}^{n} X_l\ t_b/\sqrt{t_b}\ \mathrm{rect}(f t_b)$$

$$= \frac{1}{n\sqrt{t_b}}\ \mathrm{rect}(f t_b) \sum_{l=1}^{n} X_l$$



$$\boxed{\text{The bandwidth is } 5\,MHz}$$

$$PSD = \frac{e_s}{t_b} |\mathcal{F}(u(t - l t_b))|^2$$

$$= \frac{e_s}{t_b} |\sqrt{t_b}\ \mathrm{rect}_{1/t_b}(f)|^2$$

$$= e_s\ \mathrm{rect}_{1/t_b}(f)$$

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Problem 2.2.A.5:

$r_x(t_1, t_2) = E[x(t_1)x^*(t_2)]$

$x(t) = \sum_{l=1}^{n} x_l\, u(t - lt_b)$

1) $r_x(t_1, t_2) = E[\sum_{l=1}^{n} x_l\, u(t_1 - lt_b)(\sum_{p=1}^{n} x_p\, u(t_2 - pt_b)^*]$

$= E[\sum_{l=1}^{n}\sum_{p=1}^{n} x_l x_p^*\, u(t_1 - lt_b)\, u^*(t_2 - pt_b)]$

$= \sum_{l=1}^{n}\sum_{p=1}^{n} u(t_1 - lt_b)\, u^*(t_2 - pt_b)\, E[x_l x_p^*]$

$E[x_l x_p^*] = e_s\, \delta(l-p)$ ($\emptyset$ when $p \neq l$)

$\boxed{r_x(t_1, t_2) = e_s \sum_{l=1}^{n} u(t_1 - lt_b)\, u^*(t_2 - lt_b)}$

2) $r_x(t_1, t_1) = e_s \sum_{l=1}^{n} u(t_1 - lt_b)\, u^*(t_1 - lt_b)$

If we assume a real-valued $u(t)$...

$\boxed{r_x(t_1, t_1) = e_s \sum_{l=1}^{n} (u(t_1 - lt_b))^2}$

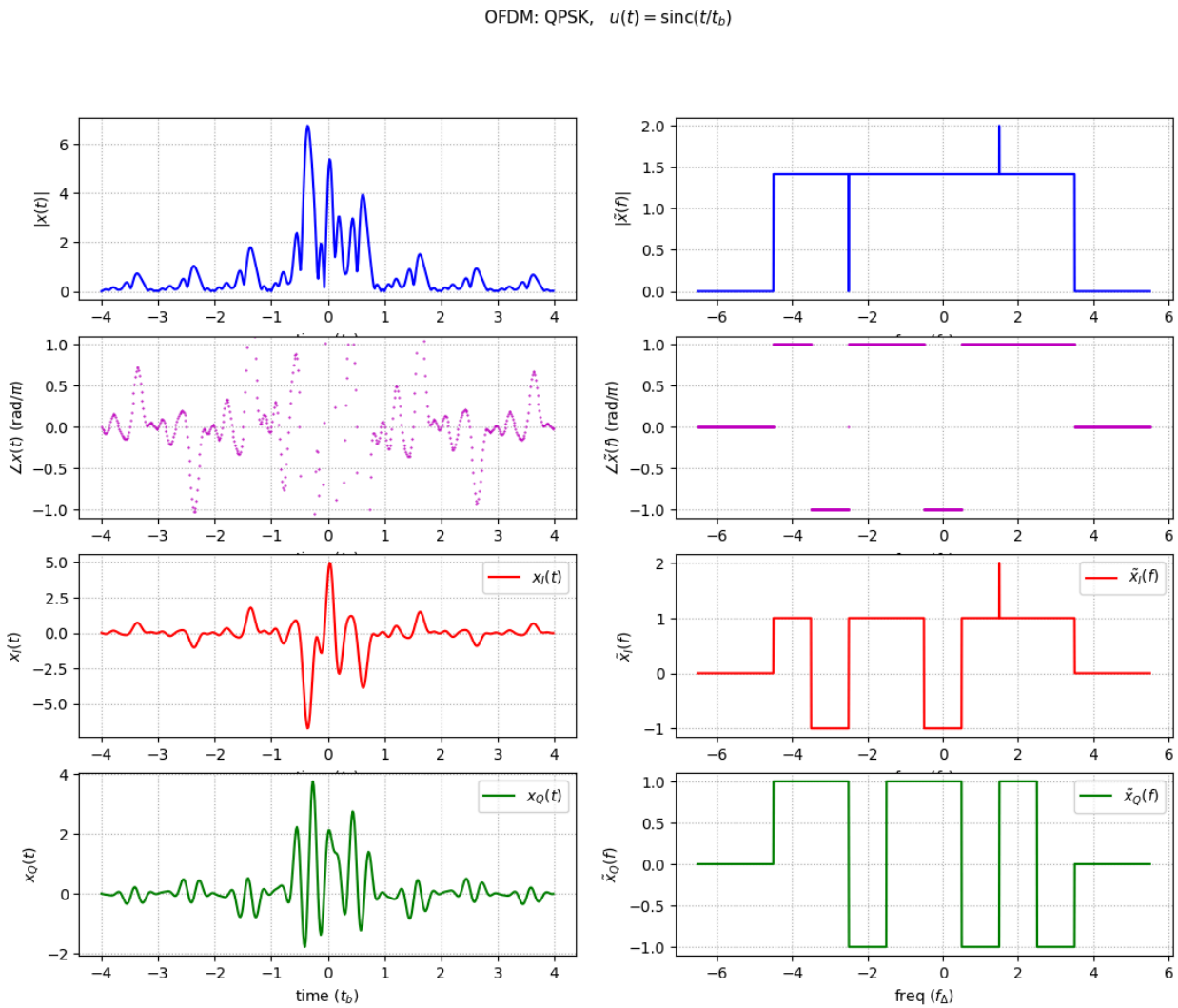
3) $r_x(t_1 + t_b, t_2 + t_b) = e_s \sum_{l=1}^{n} u(t_1 + t_b - lt_b)\, u^*(t_2 + t_b - lt_b)$

$= \boxed{e_s \sum_{l=1}^{n} u(t_1 - (l-1)t_b)\, u^*(t_2 - (l-1)t_b)}$

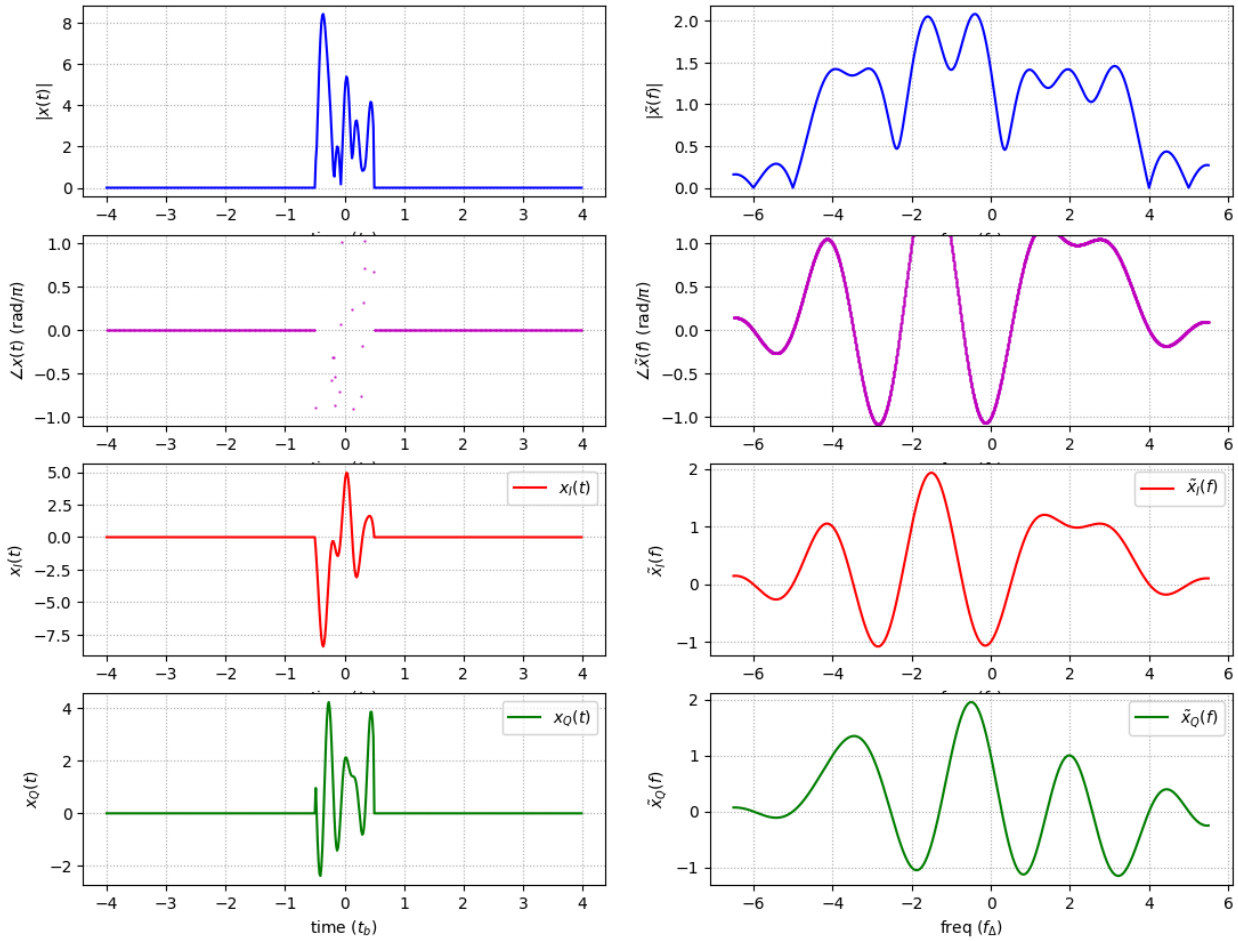
This function is equivalent to $e_s \sum_{l=0}^{n} u(t_1 - lt_b)\, u^*(t_2 - lt_b)$, so everything is the same, just shifted by 1 symbol.

Problem 2.2.A.6;7, 8 solved in python, see next pages

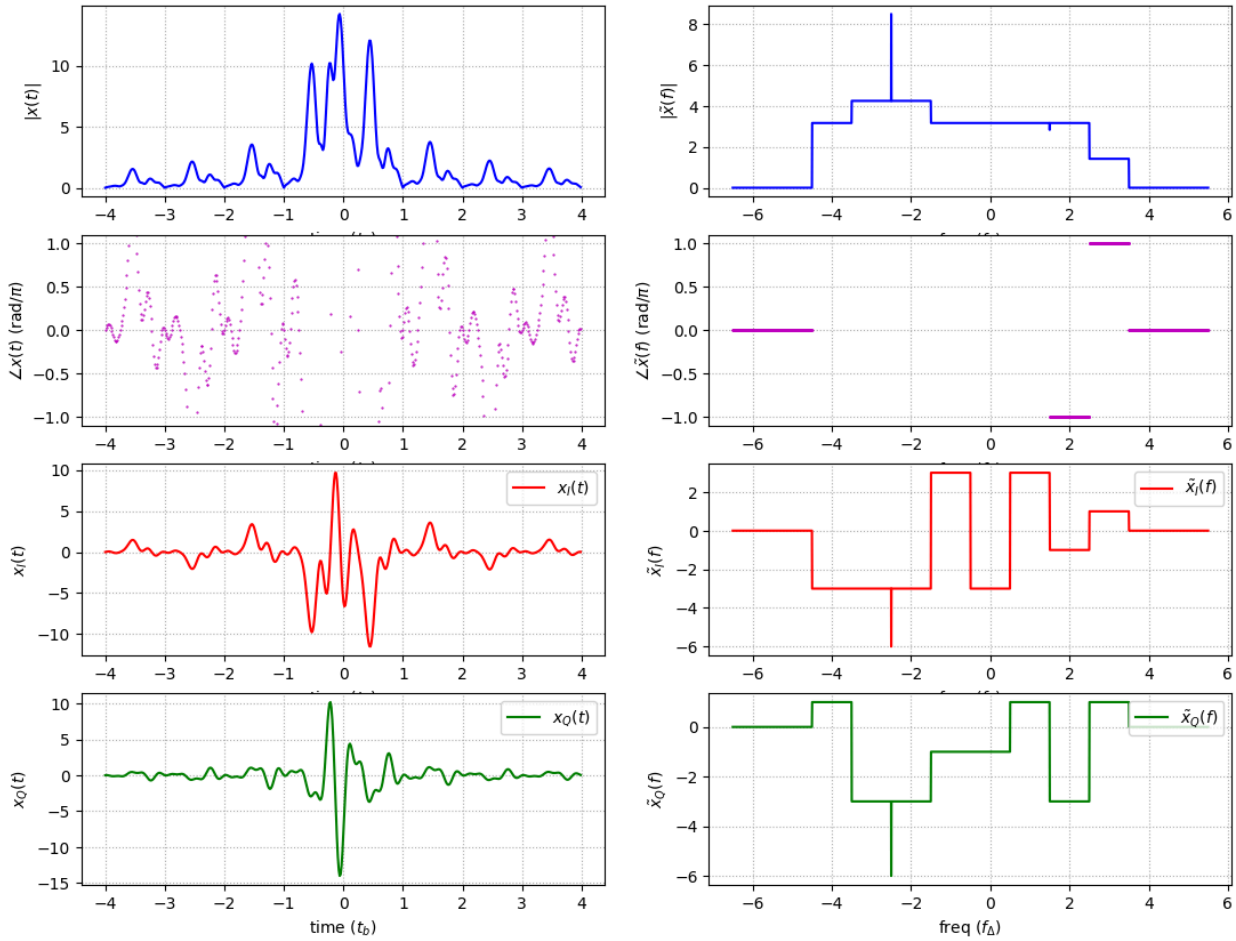
Problem 2.2.A.6



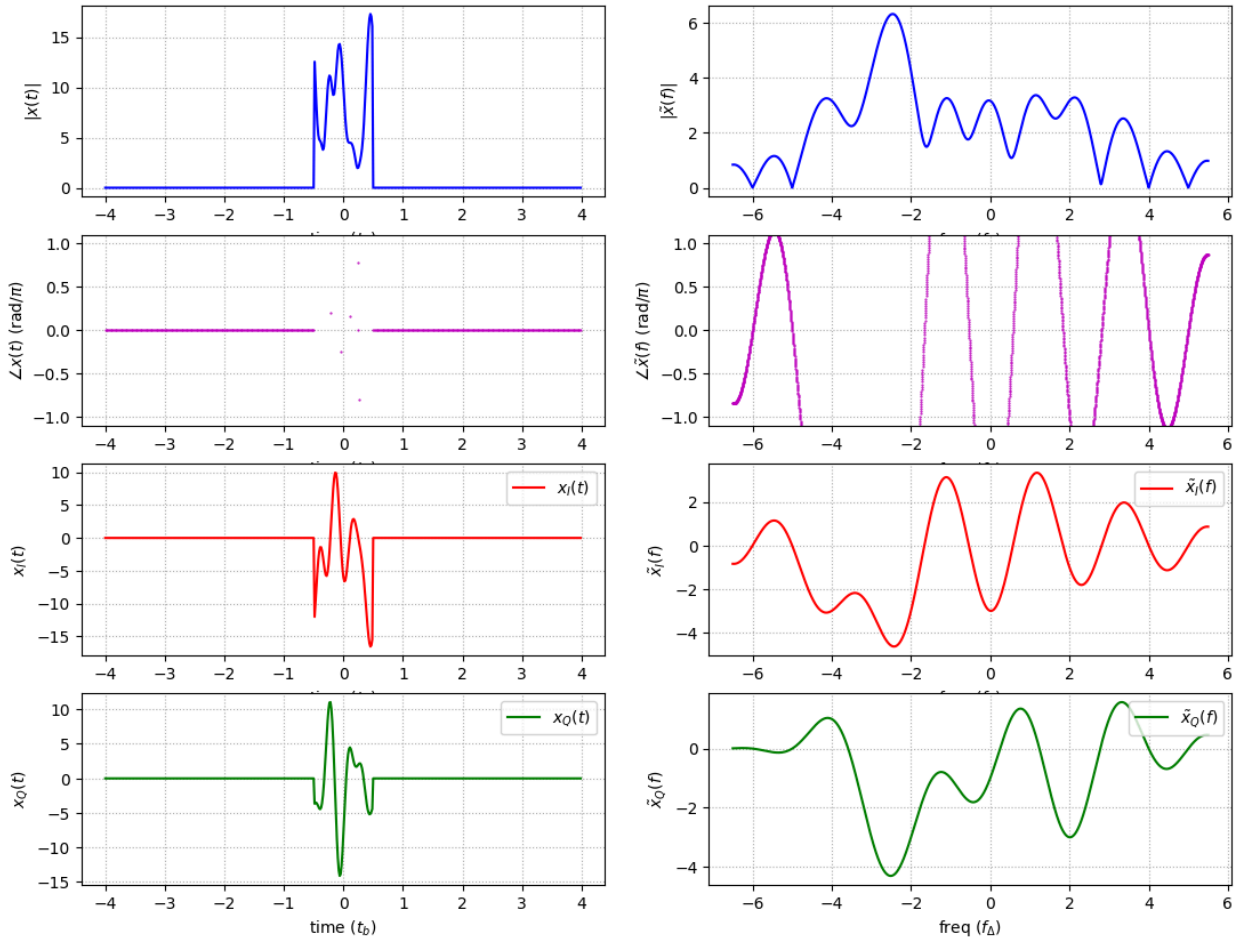
OFDM: QPSK, $u(t) = \mathrm{rect}(t/t_b)$ (square pulse)



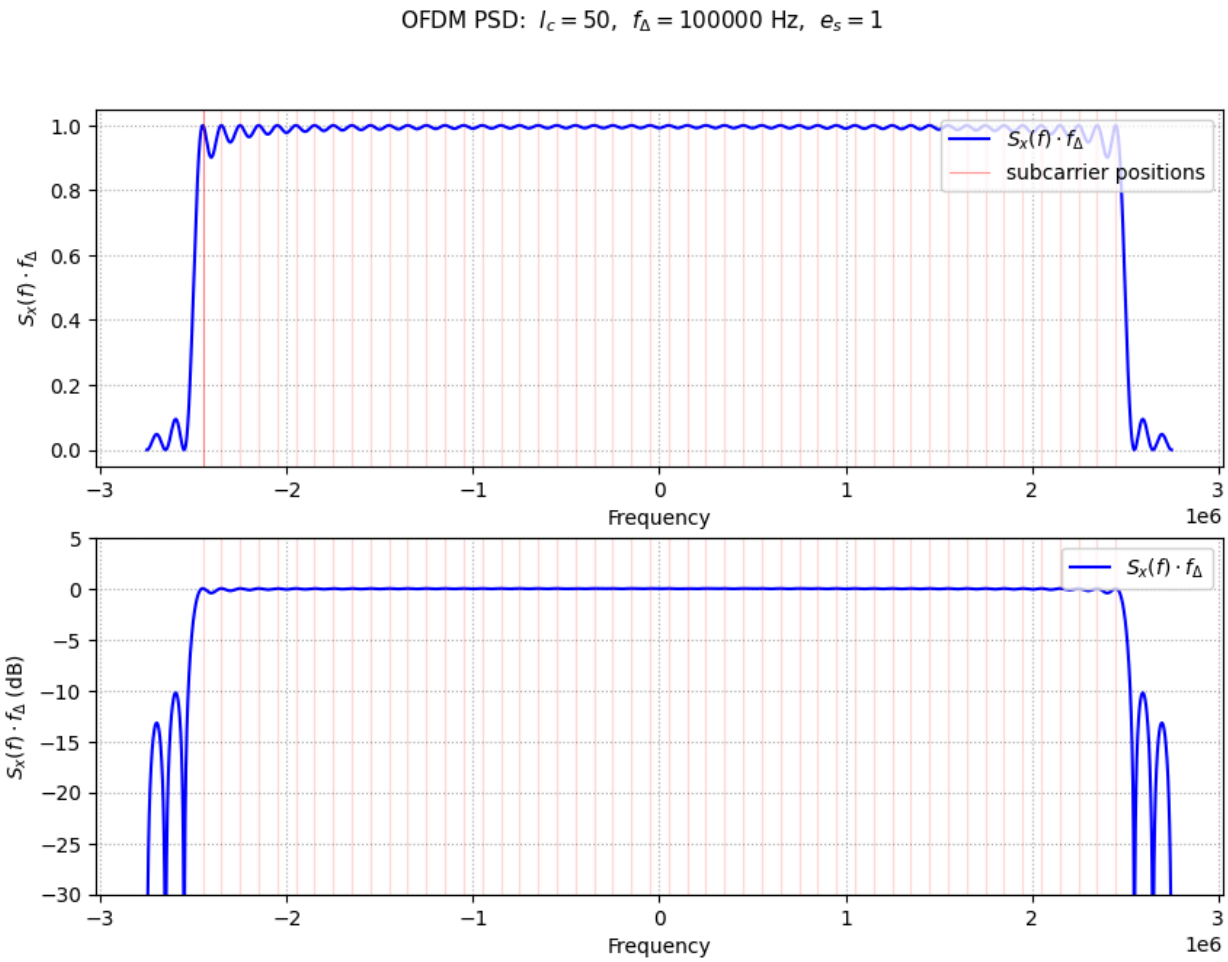
OFDM: 16-QAM, $u(t) = \mathrm{sinc}(t/t_b)$



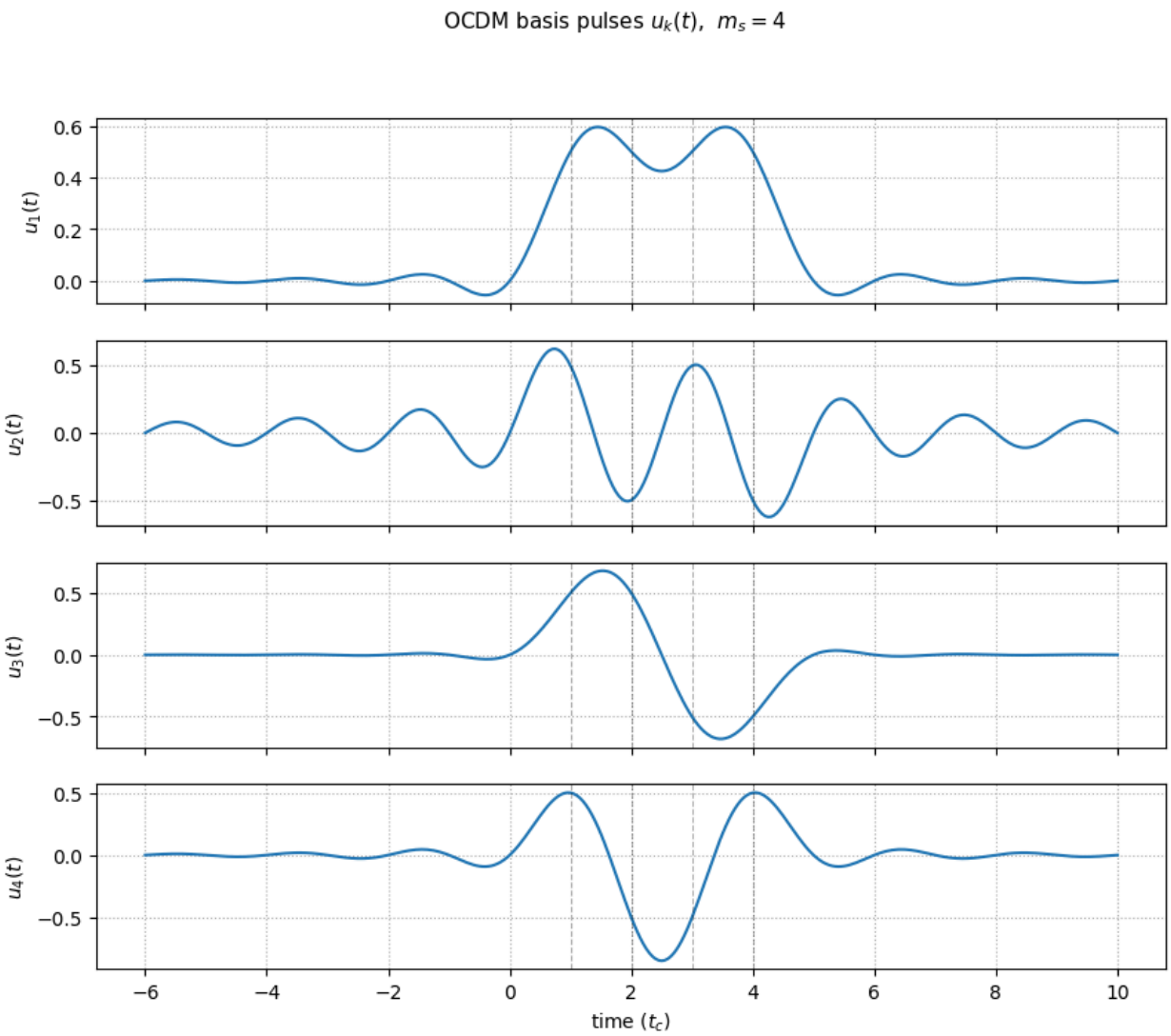
OFDM: 16-QAM, $u(t) = \mathrm{rect}(t/t_b)$ (square pulse)

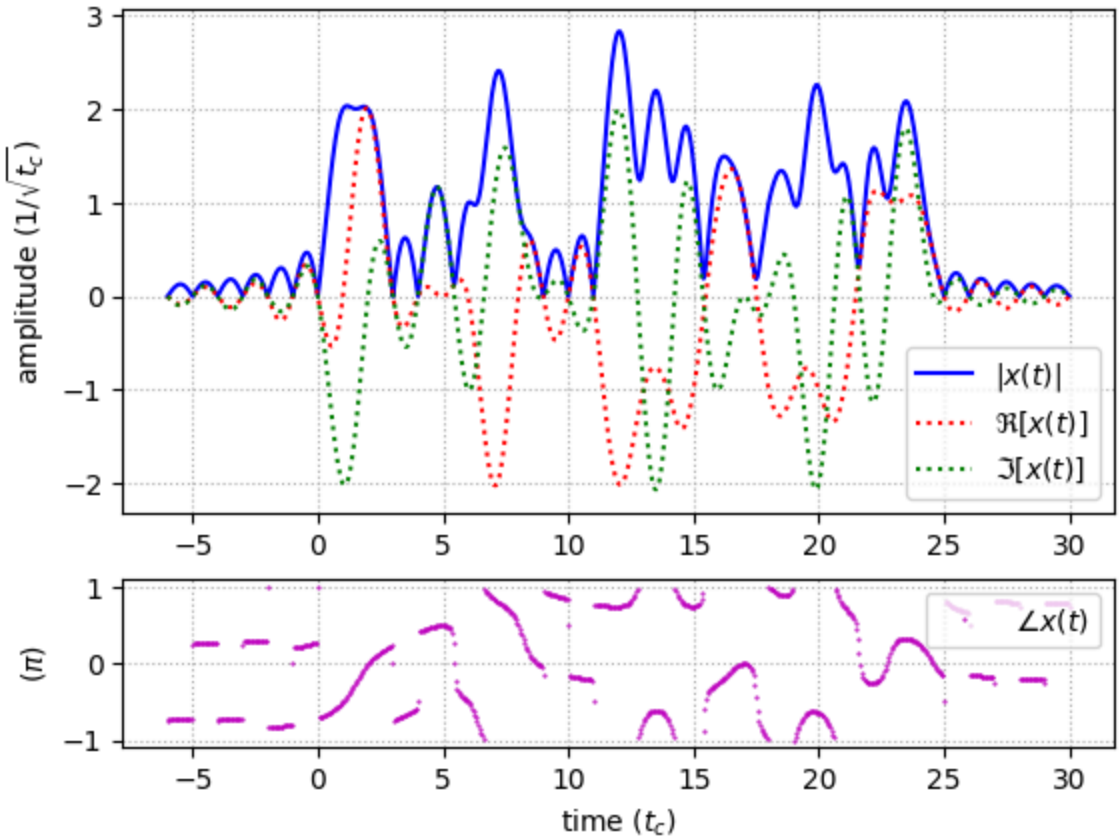


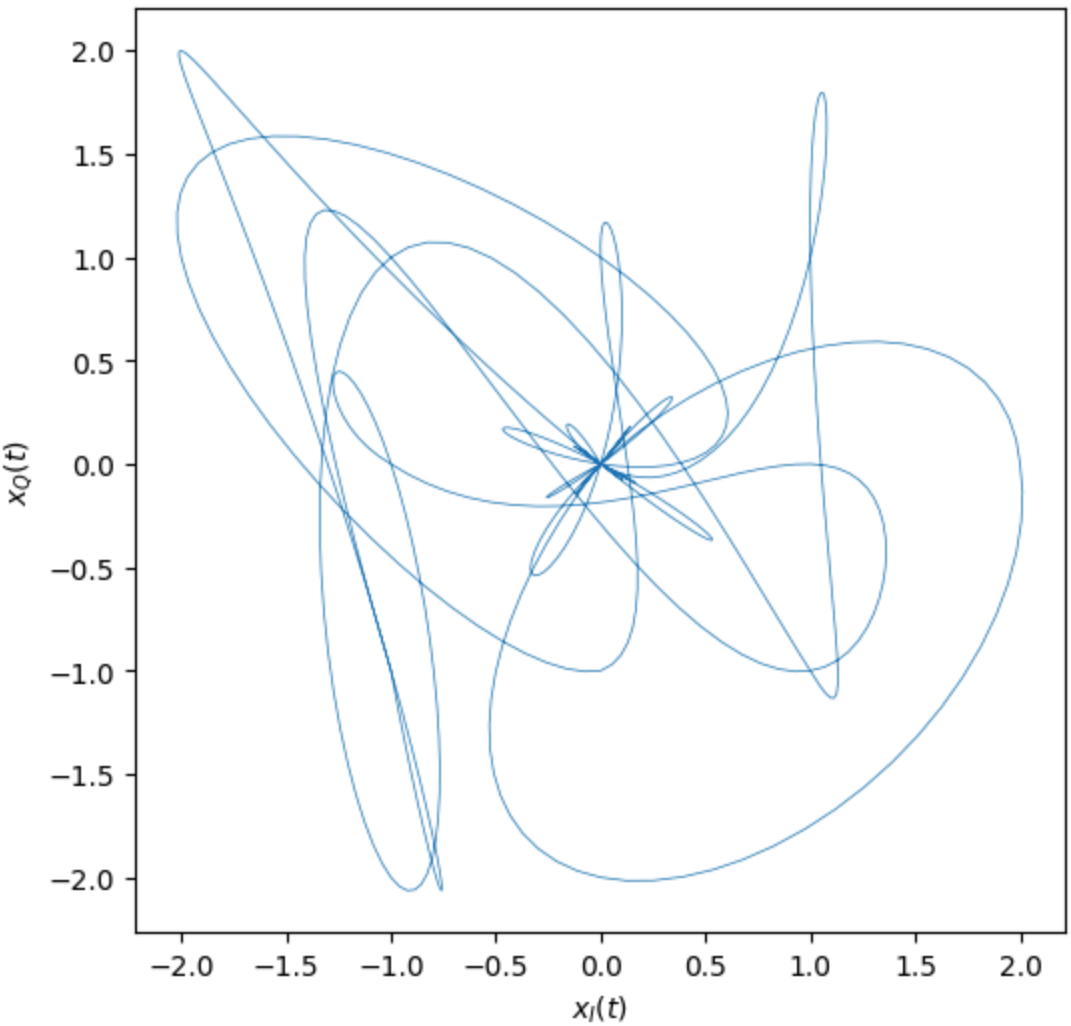
Problem 2.2.A.7

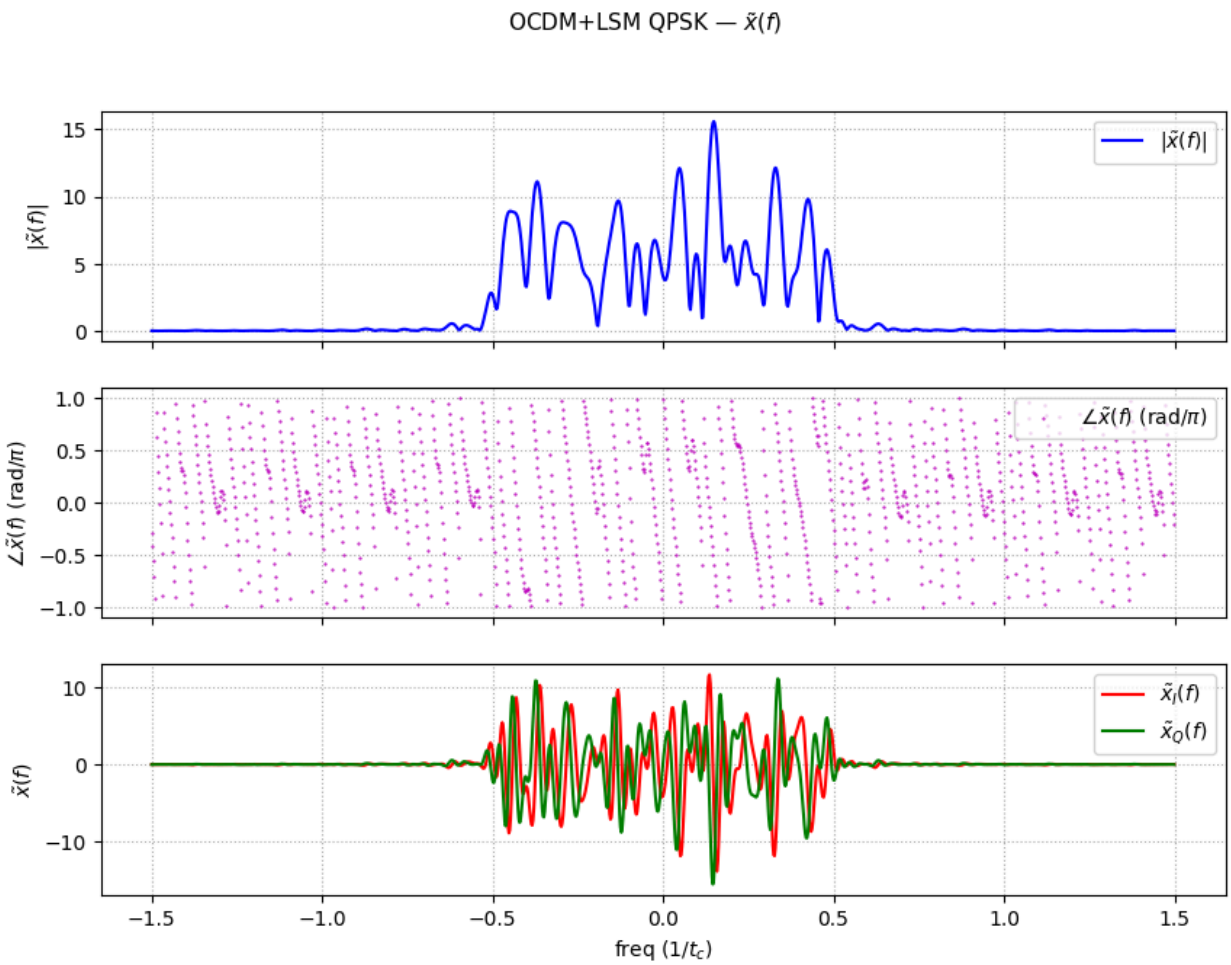


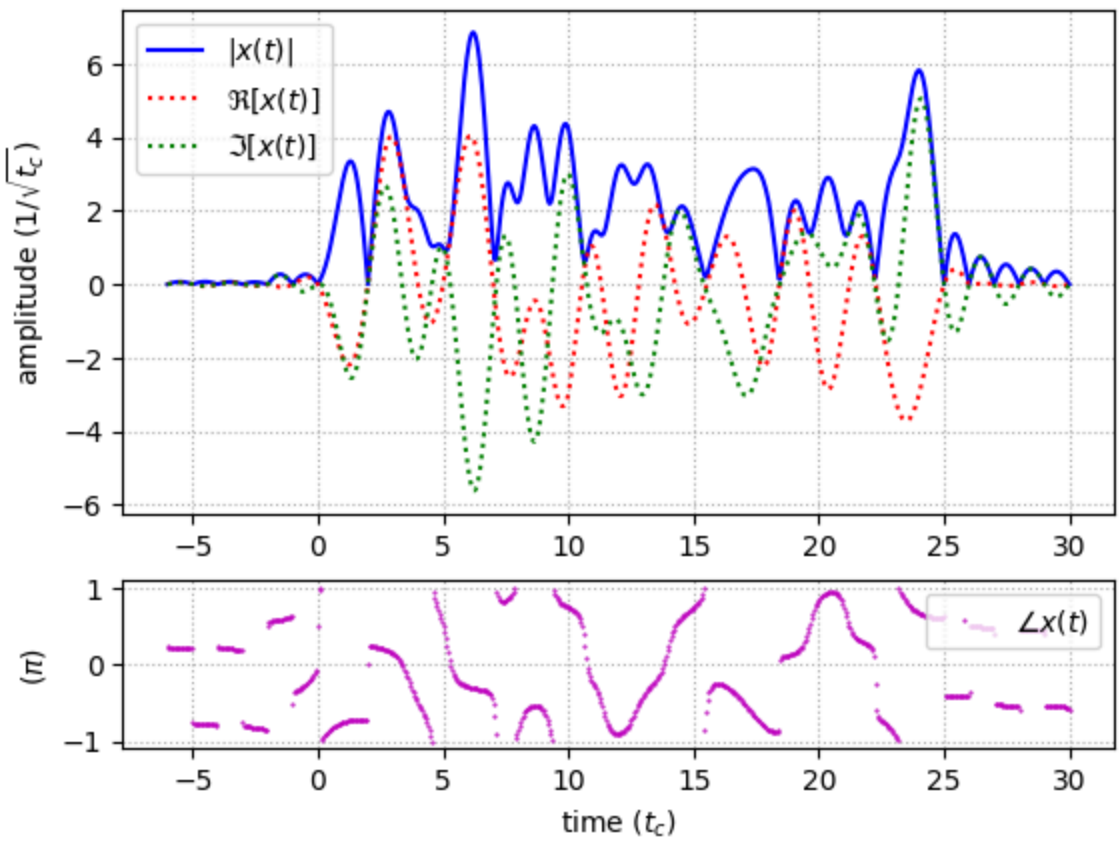
Problem 2.2.A.8

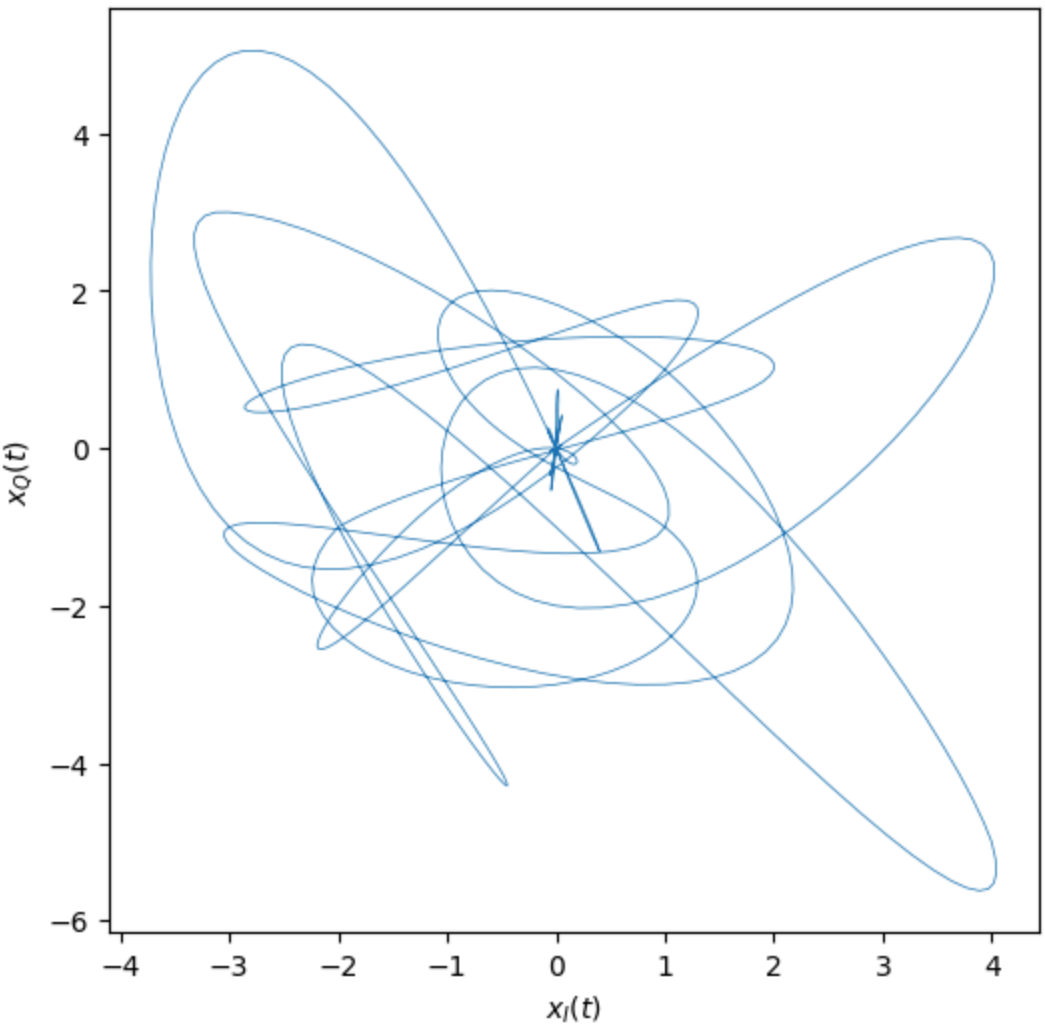

OCDM basis pulses $u_k(t)$, $m_s = 4$
$u_1(t)$
$u_2(t)$
$u_3(t)$
$u_4(t)$
0.6
0.4
0.2
0.0
0.5
0.0
−0.5
0.5
0.0
−0.5
0.5
0.0
−0.5
−6
−4
−2
0
2
4
6
8
10
time (t_c)

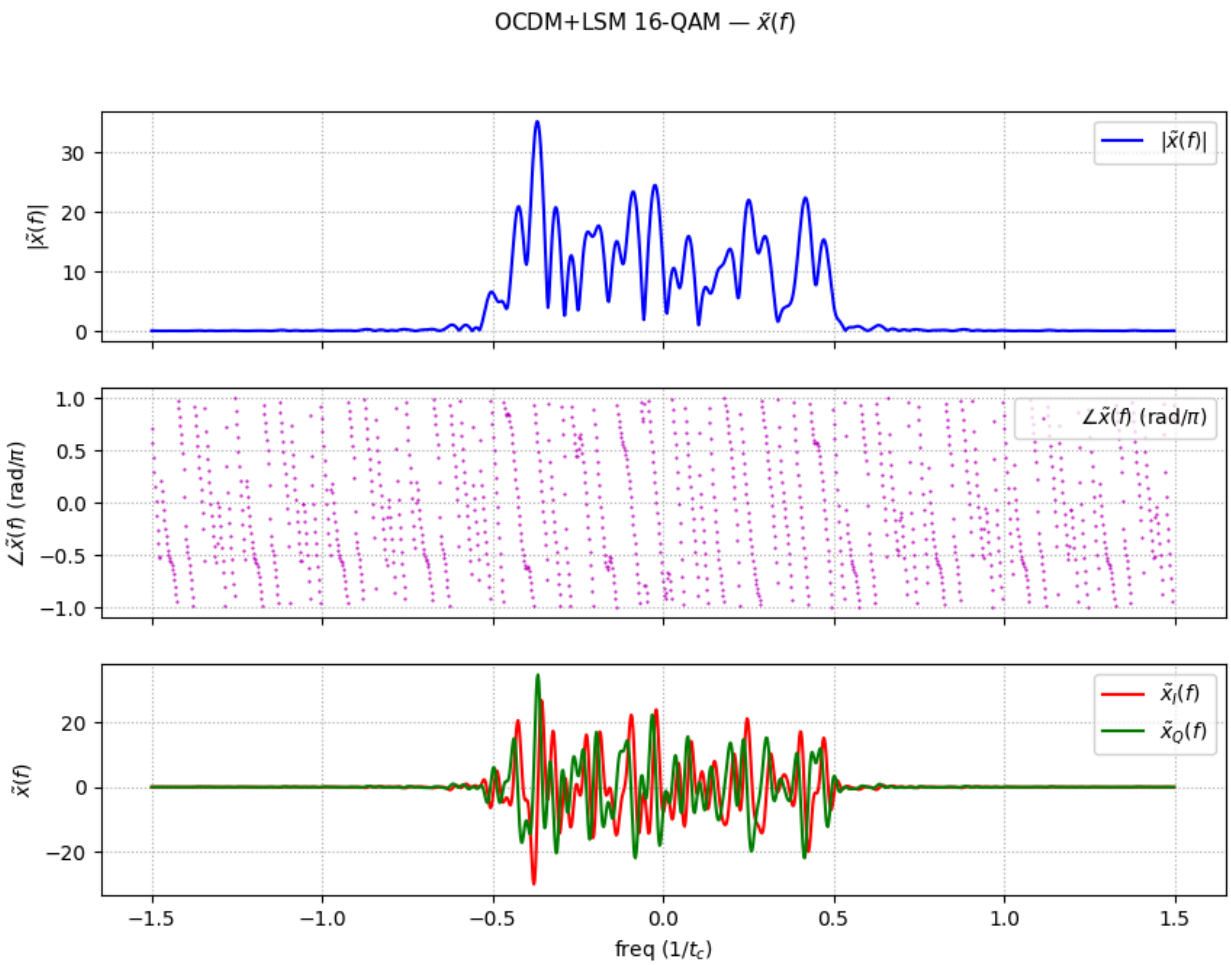

amplitude ($1/\sqrt{t_c}$)
$|x(t)|$
$\Re[x(t)]$
$\Im[x(t)]$
(π)
$\angle x(t)$
time (t_c)


2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$x_Q(t)$
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
$x_I(t)$


OCDM+LSM QPSK — $\tilde{x}(f)$
$|\tilde{x}(f)|$
$\angle\tilde{x}(f)$ (rad/π)
$\tilde{x}_I(f)$
$\tilde{x}_Q(f)$
$\tilde{x}(f)$
freq (1/t_c)


$|x(t)|$
$\Re[x(t)]$
$\Im[x(t)]$
amplitude $(1/\sqrt{t_c})$
$\angle x(t)$
(π)
time (t_c)


4
2
0
−2
−4
−6
−4
−3
−2
−1
0
1
2
3
4
$x_Q(t)$
$x_I(t)$


OCDM+LSM 16-QAM — $\tilde{x}(f)$
$|\tilde{x}(f)|$
$|\tilde{x}(f)|$
$\angle\tilde{x}(f)$ (rad/π)
$\angle\tilde{x}(f)$ (rad/π)
$\tilde{x}_I(f)$
$\tilde{x}_Q(f)$
$\tilde{x}(f)$
freq (1/t_c)